Literature-implied answer: high-dimensional Krivine schemes are asymptotically optimal

Run fa_banach_001, agent lane 17

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Source Question

The source paper is Braverman–Makarychev–Makarychev–Naor, *The Grothendieck constant is strictly smaller than Krivine's bound*, arXiv:1103.6161 [1]. After disproving Krivine's conjecture and Konig's intermediate conjecture, the authors ask whether one can analytically describe the high-dimensional maximizers

$$f^{(n)}_{\max}, g^{(n)}_{\max} : \mathbb{R}^n \to \{-1, 1\}$$

of Konig's bilinear form, and whether these maximizers form an alternating Krivine rounding scheme. They note that such a positive answer would yield a formula recovering K_G from high-dimensional operator norms.

Supporting Theorem

Naor–Regev, *Krivine schemes are optimal*, arXiv:1205.6415 [2], proves the following theorem: for every $k \in \mathbb{N}$ there is a k-dimensional oblivious Krivine scheme of quality

$$(1 + O(1/k))K_G.$$

Their introduction explicitly frames this as showing that oblivious Krivine rounding schemes capture K_G exactly in the asymptotic sense: to understand the Grothendieck constant it suffices, up to a factor tending to one, to focus on partitions of Euclidean space used in Krivine schemes.

Conclusion

This gives a literature-implied answer to the broad high-dimensional direction opened in arXiv:1103.6161: high-dimensional Krivine schemes are asymptotically optimal and lose only a $1 + O(1/k)$ factor.

The scope is partial. The later theorem does not identify the actual maximizers of Konig's bilinear form, does not give an analytic formula for the "tiger partition" f_∞, and does not prove convergence of the iteration described in the source paper. Those analytic/maximizer questions should remain open in the run memory.

References

[1] M. Braverman, K. Makarychev, Y. Makarychev, and A. Naor, *The Grothendieck constant is strictly smaller than Krivine's bound*, arXiv:1103.6161, 2011.

[2] A. Naor and O. Regev, *Krivine schemes are optimal*, arXiv:1205.6415, 2012.